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ELECTRIC CONNECTOR AND CABLE ASSEMBLY

Abstract

An electric connector includes a housing, a fixing seat and at least one latch. The housing includes a body. The body defines a space therein and has two outer surfaces opposite to each other. Each of the first outer surfaces has a platform and at least one pivot formed thereon. The fixing seat is disposed in the space and configured to mount some electrical terminals. The latch is pivotally connected with the pivots and configured to rotate relative to the body from a locking status to a releasing status through an intermediate status. Each latch is in close proximity to the platforms in the locking status, the releasing status and the intermediate status, thereby an outer edge of the latch being in contact with the platforms when the latch is on impact.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to electric connectors and cable assemblies respectively including one of these electric connectors. More particularly, the present disclosure relates to electric connectors and cable assemblies for power supply to servers.

Description of Related Art

[0002] In the US patent U.S. Pat. No. 8,979,568B2, an electric connector 1 is disclosed. The electric connector 1 includes two locking bars 8. The locking bars 8 are connected with the bearing pivots 12 of the first housing half 2, such that the locking bars 8 can be rotated relative to the first housing half 2. However, the locking bars 8 are free from contact with any other portion of the first housing half 2. Therefore, when there is an impact to either one of the locking bars 8 in some status, the force born by this locking bar 8 will all be transferred to the bearing pivots 12 connected with this locking bar 8, leading to a high risk of fracture of the bearing pivots 12.

SUMMARY

[0003] A technical aspect of the present disclosure is to provide an electric connector, which can effectively improve the durability.

[0004] According to an embodiment of the present disclosure, an electric connector includes a housing, a fixing seat and at least one latch. The housing includes a body. The body has two first outer surfaces opposite to each other. The body defines a space therein. Each of the first outer surfaces has a platform and at least one pivot formed thereon. The fixing seat is disposed in the space and configured to mount a plurality of electrical terminals. The latch is pivotally connected with the pivots and configured to rotate relative to the body from a locking status to a releasing status through an intermediate status. The latch is in close proximity to the platforms in the locking status, the releasing status and the intermediate status, thereby an outer edge of the latch being in contact with the platforms when the latch is on impact.

[0005] In one or more embodiments of the present disclosure, each of the platforms has a bottom abutting surface, a side abutting surface and a curved surface. The bottom abutting surface, the side abutting surface and the curved surface are respectively connected with a corresponding one of the first outer surfaces. The bottom abutting surface is connected with the side abutting surface via the curved surface. The latch is in close proximity to the curved surface in the locking status, the intermediate status and the releasing status.

[0006] In one or more embodiments of the present disclosure, the latch includes a handle and two baffles. The baffles are separated from each other and connected to two opposite ends of the handle. Each of the baffles has a first surface, a second surface, a third surface and a fourth surface. The first surface is opposite to the second surface and proximal to the body. The second surface is distal to the body. The third surface is perpendicularly connected with the first surface and defines a first groove for pivotally connecting with a corresponding one of the pivots. The fourth surface is away from the third surface and perpendicularly connected with the first surface. The fourth surface defines a second groove for snapping with the target connector.

[0007] In one or more embodiments of the present disclosure, the pivots extend along an axis. The latch rotates about the axis relative to the body. Each of the baffles further has a fifth surface and a sixth surface. The fifth surface is circularly curved around the axis and connected between the first surface and the second surface. The sixth surface is tangentially connected with the fifth surface and connected between the first surface and the second surface. The sixth surface is flat. The fifth surface is in close proximity to the curved surface of a corresponding one of the platforms in the locking status, the releasing status and the intermediate status, while the sixth surface is in close proximity to the bottom abutting surface of the corresponding one of the platforms in the releasing status.

[0008] In one or more embodiments of the present disclosure, the axis defines a perpendicular

distance from the platforms. Each of the fifth surfaces has a radius relative to the axis. The radius is equal to the perpendicular distance.
[0009] In one or more embodiments of the present disclosure, a profile of each of the curved surfaces and a corresponding one of the bottom abutting surfaces matches with a profile of the fifth surface and the sixth surface of a corresponding one of the baffles.
[0010] In one or more embodiments of the present disclosure, each of the curved surfaces is circularly curved around the axis.
[0011] In one or more embodiments of the present disclosure, each of the baffles further has a seventh surface. The seventh surface is curved and connected between the first surface and the second surface. The sixth surface is connected between the fifth surface and the seventh surface. The seventh surface is configured to abut against a corresponding one of the bottom abutting surfaces. The profile of each of the bottom abutting surfaces further matches with a profile of the seventh surface of a corresponding one of the baffles.
[0012] In one or more embodiments of the present disclosure, each of the platforms has a second outer surface away from the body. Each of the bottom abutting surfaces, a corresponding one of the curved surfaces and a corresponding one of the side abutting surfaces are connected between a corresponding one of the first outer surfaces and a corresponding one of the second outer surfaces. Each of the platforms further has a plurality of holes. The holes are distributed on the second outer surface.
[0013] In one or more embodiments of the present disclosure, each of the third surfaces is at least partially curved circularly around the axis.
[0014] In one or more embodiments of the present disclosure, each of the pivots is a cylindrical column.
[0015] In one or more embodiments of the present disclosure, the baffles are structurally symmetrical with each other.
[0016] In one or more embodiments of the present disclosure, the handle and the baffles are one piece formed.
[0017] In one or more embodiments of the present disclosure, the body, the platforms and the pivots are one piece formed.
[0018] In one or more embodiments of the present disclosure, a quantity of the latch is two. The latches are structurally symmetrical with each other.
[0019] According to an embodiment of the present disclosure, an electric connector includes a housing and a latch. The housing has a front surface and a back surface. The front surface has a first pivot and a protrusion formed thereon. The back surface has a second pivot aligned with the first pivot. The latch is pivotally connected with the first pivot and the second pivot. The latch is configured to rotate relative to the housing from a locking status to a releasing status through an intermediate status. The latch is in close proximity to the protrusion in the locking status, the intermediate status and the releasing status, thereby an outer contour of the latch being in contact with the protrusion when the latch is on impact in either one of the locking status, the releasing status and the intermediate status.
[0020] In one or more embodiments of the present disclosure, the protrusion has a bottom abutting surface, a side abutting surface and a curved surface connected between the bottom abutting surface and the side abutting surface. The bottom abutting surface is substantially perpendicular to the side abutting surface. When the latch is in the locking status, an outer contour of the latch is in contact with the side abutting surface. When the latch is in the releasing status, the outer contour of the latch is in contact with the bottom abutting surface. When the latch is in the intermediate status, the outer contour of the latch is away from the side abutting surface and the bottom abutting surface.
[0021] A technical aspect of the present disclosure is to provide a cable assembly, which can effectively improve the durability.

[0022] According to an embodiment of the present disclosure, a cable assembly includes a cable and one of the electric connectors mentioned above. The electric connector is electrically connected with an end of the cable.
[0023] The above-mentioned embodiments of the present disclosure have at least the following advantages: since the baffles of each of the latches are in close proximity to the platforms in all the locking status, the releasing status and the intermediate status, when there is an impact to either one of the latches in either one of the locking status, the releasing status or the intermediate status, the force born by the latch will be transferred not only to the pivots of the housing but also to the platforms through the curved surfaces (and the bottom abutting surfaces or the side abutting surface, depending on the status of this latch at the moment of impact and the angle of the force exerting on this latch). In this way, the force exerting on the pivots in an impact to the latch will be effectively reduced, and the risk of fracture of the pivots is greatly reduced. Therefore, the durability of the electric connector is effectively improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:
[0025] FIG. **1** is a schematic view of an electric connector according to an embodiment of the present disclosure;
[0026] FIG. **2** is an exploded view of the electric connector of FIG. **1**;
[0027] FIG. **3** is a side view of the electric connector of FIG. **1**;
[0028] FIG. **4** is a front view of the housing of FIG. **1**;
[0029] FIG. **5** is a sectional view along the sectional line B-B of FIG. **2**;
[0030] FIG. **6** is a schematic view of one of the latches of FIG. **1**;
[0031] FIG. **7** is a sectional view along the sectional line A-A of FIG. **1**;
[0032] FIG. **8** is a front view of the electric connector of FIG. **1**, in which the latches are in the locking status;
[0033] FIG. **9** is a front view of the electric connector of FIG. **1**, in which the latches are in the releasing status; and
[0034] FIG. **10** is a front view of the electric connector of FIG. **1**, in which the latches are in the intermediate status.
DETAILED DESCRIPTION
[0035] Drawings will be used below to disclose embodiments of the present disclosure. For the sake of clear illustration, many practical details will be explained together in the description below. However, it is appreciated that the practical details should not be used to limit the claimed scope. In other words, in some embodiments of the present disclosure, the practical details are not essential. Moreover, for the sake of drawing simplification, some customary structures and elements in the drawings will be schematically shown in a simplified way. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.
[0036] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.
[0037] FIGS. **1-10** are drawn according to the actual scale. For the sake of concise description in this specification, the scales of the elements are not listed one by one. However, the scale and

position of each element should be regarded as part of the scope of this specification.

[0038] Furthermore, if the description in this specification is insufficient to explain the detailed design of the elements and makes it uneasy to implement, please take the electrical plug-in connector disclosed in the U.S. Pat. No. 8,979,568B2 as a reference.

[0039] As shown in FIGS. **1-10**, a cable assembly with an electric connector **100**, two of which are connected at both ends of a cable **160** respectively, is provided in the present disclosure. Only one end of the cable assembly is showed for clarity consideration. Moreover, for example, the electric connector **100** is applied to connect with a target connector **200** (please see FIGS. **8-10** for the target connector **200**) of a server for the sake of high current power supply.

[0040] FIG. **1** is a schematic view of an electric connector **100** according to an embodiment of the present disclosure. FIG. **2** is an exploded view of the electric connector **100** of FIG. **1**. In this embodiment, as shown in FIGS. **1-2**, the electric connector **100** includes a housing **110**, a fixing seat **120**, two latches **130**, a plurality of electrical terminals **140** and a molding piece **150** (please see FIGS. **8-10** for the molding piece **150**). The latches **130** are structurally symmetrical with each other.

[0041] In this embodiment, the electric connector **100** is a plug end connector with a cable **160** connected therewith. However, while the electric connector **100** is a socket end connector without cable connected therewith, the molding piece **150** and the cable **160** might be omitted.

[0042] FIG. **3** is a side view of the electric connector **100** of FIG. **1**. FIG. **4** is a front view of the housing **110** of FIG. **1**. As shown in FIGS. **1-4**, the housing **110** includes a body **111**, two platforms **112** and two pivot sets **113**.

[0043] The body **111** is hollow-shaped and has two first outer surfaces **111**S opposite to each other. The body **111** defines a space SP therein.

[0044] Both of the platforms **112** are respectively disposed on a corresponding one of the first outer surfaces **111**S. Both of the pivot sets **113** are disposed on the first outer surfaces **111**S along two axes XL separated from each other in a parallel manner. All of the body **111**, the platforms **112** and the pivot sets **113** are one piece formed of insulating material by injection molding process.

[0045] Furthermore, as shown in FIG. **2**, each of the pivot sets **113** includes two pivots **1131**. The pivots **1131** are separated from each other and disposed on the first outer surfaces **111**S along a corresponding one of the axes XL. In this embodiment, for example, each of the pivots **1131** is substantially cylindrical-shaped column.

[0046] FIG. **5** is a sectional view along the sectional line B-B of FIG. **2**. FIG. **6** is a schematic view of one of the latches **130** of FIG. **1**. As shown in FIGS. **1-3** and **5-6**, each of the latches **130** includes a handle **131** and two baffles **132**. The baffles **132** are separated from each other and connected to two opposite ends of the handle **131**. The baffles **132** are structurally symmetrical with each other. The handle **131** and the baffles **132** are one piece formed. Each of the baffles **132** has a first surface **132S1**, a second surface **132**S**2**, a third surface **132**S**3**, a fourth surface **132S4**, a fifth surface **132**S**5**, a sixth surface **132**S**6** and a seventh surface **132**S**7**.

[0047] As shown in FIGS. **1-3** and **5-6**, the first surface **132S1** is opposite to the second surface **132**S**2** and proximal to the body **111** of the housing **110**. The second surface **132S2** is distal to the body **111** of the housing **110**. The third surface **132**S**3** of each of the baffles **132** is perpendicularly connected with the first surface **132S1** and defines a first groove G**1**. In other words, the first groove G**1** is located on the first surface **132S1**. Furthermore, each of the third surfaces **132S3** is at least partially curved circularly around a corresponding one of the axes XL, such that the first groove G**1** is at least partially circular in shape. The fourth surface **132S4** is away from the third surface **132**S**3** and perpendicularly connected with the first surface **132S1**. The fourth surface **132**S**4** defines a second groove G**2**. In other words, the second groove G**2** is also located on the first surface **132S1** but is separated from the first groove G**1**.

[0048] Furthermore, as shown in FIGS. **5-6**, the fifth surface **132**S**5** is circularly curved around the axis XL and connected between the first surface **132S1** and the second surface **132S2**. The sixth surface **132**S**6** is tangentially connected with the fifth surface **132**S**5** and connected between the first surface **132**S**1** and the second surface **132**S**2**. In this embodiment, the sixth surface **132**S**6** is a flat surface.

[0049] On the other hand, as shown in FIG. **4**, each of the axes XL defines a perpendicular distance D from the platforms **112**. Furthermore, as shown in FIGS. **1-4**, each of the platforms **112** is T-shaped. Each of the platforms **112** has a bottom abutting surface **112**A, a side abutting surface **112**C and a curved surface **112**B respectively on both of the two sides of the T-shaped structure. Each of the bottom abutting surfaces **112**A connects with a corresponding one of the side abutting surfaces **112**C via a corresponding one of the curved surfaces **112**B. Each of the bottom abutting surfaces **112**A is substantially perpendicular to a corresponding one of the side abutting surface **112**C.

[0050] Each of the bottom abutting surfaces **112**A is respectively connected with a corresponding one of the first outer surfaces **111**S. To be specific, as shown in FIG. **4**, each of the axes XL defines the perpendicular distance D from the corresponding one of the curved surfaces **112**B of each of the platforms **112**. Correspondingly, as shown in FIG. **5**, each of the fifth surfaces **132**S**5** has a radius R relative to a corresponding one of the axes XL. In this embodiment, the radius R is substantially equal to the perpendicular distance D.

[0051] In addition, as shown in FIGS. **5-6**, the seventh surface **132**S**7** is curved and connected between the first surface **132**S**1** and the second surface **132**S**2**. The sixth surface **132**S**6** is connected between the fifth surface **132**S**5** and the seventh surface **132**S**7**. It is worth to note that, in this embodiment, a profile of each of the bottom abutting surfaces **112**A and a corresponding one of the curved surfaces **112**B matches with a profile of the fifth surface **132**S**5**, the sixth surface **132**S**6** and the seventh surface **132**S**7** of a corresponding one of the baffles **132**. To be specific, each of the curved surfaces **112**B is circularly curved around a corresponding one of the axes XL.

[0052] As shown in FIGS. **1-2** and **4**, each of the platforms **112** has a second outer surface **112**S away from the body **111**. Each of the bottom abutting surfaces **112**A, a corresponding one of the curved surfaces **112**B and a corresponding one of the side abutting surfaces **112**B are connected between a corresponding one of the first outer surfaces **111**S and a corresponding one of the second outer surfaces **112**S. Each of the platforms **112** further has a plurality of holes H. The holes H are distributed on the second outer surface **112**S.

[0053] FIG. **7** is a sectional view along the sectional line A-A of FIG. **1**. The fixing seat **120** is disposed in the space SP of the body **111** and configured to mount the electrical terminals **140**.

[0054] FIG. **8** is a front view of the electric connector **100** of FIG. **1**, in which the latches **130** are in the locking status S**1**. As shown in FIG. **1** and FIG. **8**, each of the latches **130** is pivotally connected with a corresponding one of the pivot sets **113** (please refer to FIG. **2** and FIG. **4** for the pivot sets **113**) and configured to rotate relative to the body **111** of the housing **110** about a corresponding one of the axes XL. To be specific, each of the pivots **1131** of the housing **110** is pivotally accommodated in a corresponding one of the first grooves G**1** (please see FIG. **2** and FIGS. **5-6** for the first grooves G**1**). In other words, the first groove G**1** on each of the baffles **132** is pivotally connected with a corresponding one of the pivots **1131** (please see FIG. **2** and FIG. **4** for the pivots **1131**). Moreover, as shown in FIG. **8**, both of the latches **130** are in the locking status S**1**, in which the latches **130** are rotated towards each other for engaging with a target connector **200** (presented in hidden lines). Meanwhile, the electric connector **100** is connected with a connecting cable **160** (presented in hidden lines), such that the electric connector **100** is connected between the target connector **200** and the connecting cable **160**. Furthermore, to be specific, the second groove G**2** on each of the baffles **132** is snapped with a corresponding one of the pivots (not shown) of the target connector **200**, such that the target connector **200** is connected to the electric connector **100** in a secured manner. It is worth to note that, in this embodiment, the curved surfaces **112**B are respectively configured to abut against a corresponding one of the baffles **132**. To be specific, the fifth surface **132**S**5** of each of the baffles **132** is in close proximity to a corresponding one of the curved surfaces **112**B in the locking status S**1**.

[0055] FIG. **9** is a front view of the electric connector **100** of FIG. **1**, in which the latches **130** are in the releasing status S**2**. In this embodiment, as shown in FIG. **9**, both of the latches **130** are respectively rotated from the locking status S**1** to the releasing status S**2** relative to the housing **110** about a corresponding one of the axes XL. To be specific, in the releasing status S**2**, the latches **130** are rotated away from each other, such that the electric connector **100** can be released from the target connector **200**. It is worth to note that, in this embodiment, each of the fifth surfaces **132**S**5** and the adjacent ones of the sixth surface **132**S**6** and seventh surface **132**S**7** are in close proximity to a corresponding one of the bottom abutting surfaces **112**A and a corresponding one of the curved surfaces **112**B in the releasing status S**2**.
[0056] FIG. **10** is a front view of the electric connector **100** of FIG. **1**, in which the latches **130** are in the intermediate status S**3**. In this embodiment, as shown in FIG. **10**, both of the latches **130** are respectively further rotated from the releasing status S**2** to the intermediate status S**3** relative to the housing **110** about a corresponding one of the axes XL. It is worth to note that, in this embodiment, each of the fifth surfaces **132**S**5** is still in close proximity to a corresponding one of the curved surfaces **112**B in the releasing status S**2**. Moreover, in the intermediate status S**3**, the outer contour of each of the latches **130** is away from the side abutting surface **112**C and the bottom abutting surface **112**A.
[0057] It is important to clarify that, each of the latches **130** may, but not necessarily, be always in contact and interfering with the platforms **120**. Instead, in the normal operational states of the latches and pivots, there can optionally be a small gap due to tolerances or design between the each of the latches **130** and the platforms **120**, while the said gap has to be smaller than or equal to the sum of the maximum movable range of the latch **130** on impact, the maximum elastic deformation of this latch **130**, and the maximum elastic deformation of the pivot **1131**. Moreover, as an example, the said gap is typically not greater than one-fifth of the diameter of the pivot **1131**. More specifically, the said gap is suggested to be smaller than 500 micrometer and is preferably smaller than 150 micrometers. By the said design, when the connector **100** is dropped from a height, for example, of one meter and one of the latches **130** is impacted by the floor, this latch **130**, due to displacement, deformation or any other matters, will be in contact with the platforms **120** no matter the connector **100** is in either one of the locking status, the intermediate status and the locking status before the corresponding pivots **1131** undergo irreversible plastic deformation/destruction, thus avoiding the pivots **1131** from being damaged or at least reducing the extent of damage to the pivots **1131**.
[0058] As mentioned above, since the baffles **132** of each of the latches **130** are in close proximity to the platforms **112** in all the locking status S**1**, the releasing status S**2** and the intermediate status S**3**, when there is an impact to either one of the latches **130** in either one of the locking status S**1**, the releasing status S**2** or the intermediate status S**3**, the force born by the latch **130** will be transferred not only to the pivots **1131** of the housing **110** but also to the platforms **112** through the curved surfaces **112**B (and the bottom abutting surfaces **112**A or the side abutting surfaces **112**C, depending on the status of this latch **130** at the moment of impact and the angle of the force exerting on this latch **130**). In this way, the force exerting on the pivots **1131** in an impact to the latch **130** will be effectively reduced, and the risk of fracture of the pivots **1131** is greatly reduced. Therefore, the durability of the electric connector **100** is effectively improved.
[0059] In conclusion, the aforementioned embodiments of the present disclosure have at least the following advantages: since the baffles of each of the latches are in close proximity to the platforms in all the locking status, the locking status and the intermediate status, when there is an impact to either one of the latches in either one of the locking status, the locking status or the intermediate status, the force born by the latch will be transferred not only to the pivots of the housing but also to the platforms through the curved surfaces (and the bottom abutting surfaces or the side abutting surface, depending on the status of this latch at the moment of impact and the angle of the force exerting on this latch). In this way, the force exerting on the pivots in an impact to the latch will be

effectively reduced, and the risk of fracture of the pivots is greatly reduced. Therefore, the durability of the electric connector is effectively improved.

[0060] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0061] It will be apparent to the person having ordinary skill in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of the present disclosure provided they fall within the scope of the following claims.

Claims

1. An electric connector, comprising: a housing comprising a body, the body having two first outer surfaces opposite to each other, the body defining a space therein, each of the first outer surfaces having a platform and at least one pivot formed thereon; a fixing seat disposed in the space and configured to mount a plurality of electrical terminals; and at least one latch pivotally connecting with the pivots and configured to rotate relative to the body from a locking status to a releasing status through an intermediate status, wherein the latch is in close proximity to the platforms in the locking status, the intermediate status and the releasing status, thereby an outer edge of the latch being in contact with the platforms when the latch is on impact.

2. The electric connector of claim 1, wherein each of the platforms has a bottom abutting surface, a side abutting surface and a curved surface, the bottom abutting surface, the side abutting surface and the curved surface are respectively connected with a corresponding one of the first outer surfaces, the bottom abutting surface is connected with the side abutting surface via the curved surface, the latch is in close proximity to the curved surface in the locking status, the intermediate status and the releasing status.

3. The electric connector of claim 2, wherein the latch comprises: a handle; and two baffles separated from each other and connected to two opposite ends of the handle, each of the baffles has a first surface, a second surface, a third surface and a fourth surface, the first surface is opposite to the second surface and proximal to the body, the second surface is distal to the body, the third surface is perpendicularly connected with the first surface and defines a first groove for pivotally connecting with a corresponding one of the pivots, the fourth surface is away from the third surface and perpendicularly connected with the first surface, the fourth surface defines a second groove for snapping with a target connector.

4. The electric connector of claim 3, wherein the pivots extend along an axis, the latch rotates about the axis relative to the body, each of the baffles further has a fifth surface and a sixth surface, the fifth surface is circularly curved around the axis and connected between the first surface and the second surface, the sixth surface is tangentially connected with the fifth surface and connected between the first surface and the second surface, the sixth surface is flat, wherein the fifth surface is in close proximity to the curved surface of a corresponding one of the platforms in the locking status, the releasing status and the intermediate status, while the sixth surface is in close proximity to the bottom abutting surface of the corresponding one of the platforms in the releasing status.

5. The electric connector of claim 4, wherein the axis defines a perpendicular distance from the platforms, each of the fifth surfaces has a radius relative to the axis, the radius is substantially equal to the perpendicular distance.

6. The electric connector of claim 4, wherein a profile of each of the curved surfaces and a corresponding one of the bottom abutting surfaces matches with a profile of the fifth surface and the sixth surface of a corresponding one of the baffles.

7. The electric connector of claim 6, wherein each of the curved surfaces is circularly curved around the axis.

8. The electric connector of claim 6, wherein each of the baffles further has a seventh surface, the seventh surface is curved and connected between the first surface and the second surface, the sixth surface is connected between the fifth surface and the seventh surface, the seventh surface is configured to abut against a corresponding one of the bottom abutting surfaces, the profile of each of the bottom abutting surfaces further matches with a profile of the seventh surface of a corresponding one of the baffles.

9. The electric connector of claim 6, wherein each of the platforms has a second outer surface away from the body, each of the bottom abutting surfaces, a corresponding one of the curved surfaces and a corresponding one of the side abutting surfaces are connected between a corresponding one of the first outer surfaces and a corresponding one of the second outer surfaces, each of the platforms further has a plurality of holes distributed on the second outer surface.

10. The electric connector of claim 3, wherein each of the third surfaces is at least partially curved circularly around the axis.

11. The electric connector of claim 10, wherein each of the pivots is a cylindrical column.

12. The electric connector of claim 3, wherein the baffles are structurally symmetrical with each other.

13. The electric connector of claim 3, wherein the handle and the baffles are one piece formed.

14. The electric connector of claim 1, wherein the body, the platforms and the pivots are one piece formed.

15. The electric connector of claim 1, wherein a quantity of the latch is two, the latches are structurally symmetrical with each other.

16. An electric connector, comprising: a housing having a front surface and a back surface, the front surface having a first pivot and a protrusion formed thereon, the back surface having a second pivot aligned with the first pivot; and a latch pivotally connected with the first pivot and the second pivot, the latch being configured to rotate relative to the housing from a locking status to a releasing status through an intermediate status, wherein, the latch is in close proximity to the protrusion in the locking status, the intermediate status and the releasing status, thereby an outer contour of the latch being in contact with the protrusion when the latch is on impact in either one of the locking status, the releasing status and the intermediate status.

17. The electric connector of claim 16, wherein the protrusion has a bottom abutting surface, a side abutting surface and a curved surface connected between the bottom abutting surface and the side abutting surface, the bottom abutting surface is substantially perpendicular to the side abutting surface, wherein, when the latch is in the locking status, an outer contour of the latch is in contact with the side abutting surface, when the latch is in the releasing status, the outer contour of the latch is in contact with the bottom abutting surface, when the latch is in the intermediate status, the outer contour of the latch is away from the side abutting surface and the bottom abutting surface.

18. A cable assembly, comprising: a cable; and the electric connector of claim 16, the electric connector being electrically connected with an end of the cable.